ANSWER KEY & MARKING SCHEME SET 1

Chapter 11: **World of Algorithms** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 9	Pairs: $(360,156) \to (204,156) \to (156,48) \to (108,48) \to (60,48) \to (48,12) \to (36,12) \to (24,12) \to (12,12) \to (12,0)$.
2	**(d)** 9	Pairs: $(420,364) \to (364,56) \to (308,56) \to (252,56) \to (196,56) \to (140,56) \to (84,56) \to (56,28) \to (28,28) \to (28,0)$.
3	**(a)** 23	Multiples of 3: 16; of 5: 10; of both: 3.
4	**(d)** 18	The algorithm adds the digits.
5	**(c)** 100	LCM $= \dfrac{20 \times 25}{5}$.
6	**(d)** 19	$\lfloor \sqrt{373} \rfloor = 19$.
7	**(a)** 9	Pairs: $(312,200) \to (200,112) \to (112,88) \to (88,24) \to (64,24) \to (40,24) \to (24,16) \to (16,8) \to (8,8) \to (8,0)$.
8	**(b)** 3	Pairs: $(453,384) \to (384,69) \to (315,69) \to (246,69) \to (177,69) \to (108,69) \to (69,39) \to (39,30) \to (30,9) \to (21,9) \to (12,9) \to (9,3) \to (6,3) \to (3,3) \to (3,0)$.
9	**(d)** 6	Pairs: $(294,245) \to (245,49) \to (196,49) \to (147,49) \to (98,49) \to (49,49) \to (49,0)$.
10	**(d)** 9	Pairs: $(448,176) \to (272,176) \to (176,96) \to (96,80) \to (80,16) \to (64,16) \to (48,16) \to (32,16) \to (16,16) \to (16,0)$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$(106,44) \to (62,44) \to (44,18) \to (26,18) \to (18,8) \to (10,8) \to (8,2) \to (6,2) \to (4,2) \to (2,2) \to (2,0)$; gcd $= 2$ (1); LCM $= \dfrac{106 \times 44}{2} = 2332$ (1).	1+1
12	$\lfloor \sqrt{134} \rfloor = 11$, so check $2, \ldots, 11$; The smallest divisor found is 2 since $134 = 2 \times 67$ (1); 134 is composite (1).	1+1
13	Replace (m,n) by $(n, m-n)$ re-ordered so the larger is first; $(224,154) \to (154,70) \to (84,70) \to (70,14) \to (56,14) \to (42,14) \to (28,14) \to (14,14) \to (14,0)$ (1); 8 subtractions, gcd $= 14$ (1).	1+1
14	$(111,74) \to (74,37) \to (37,37) \to (37,0)$; gcd $= 37$ (1); LCM $= \dfrac{111 \times 74}{37} = 222$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	divisors(84) $= [1,2,3,4,6,7,12,14,21,28,42,84]$, divisors(68) $= [1,2,4,17,34,68]$; Greatest common divisor from the lists $= 4$ (1); Subtraction: $(84,68) \to (68,16) \to (52,16) \to (36,16) \to (20,16) \to (16,4) \to (12,4) \to (8,4) \to (4,4) \to (4,0)$, 9 subtractions (1); gcd $= 4$ by both methods (1).	1+1+1
16	Multiples of 3: 8; multiples of 5: 5 (1); Multiples of both (counted twice): 1 (1); Count $= 8+5-1 = 12$ (1).	1+1+1
17	Rule: new $x = 2x - 4$ (1); Values: 23, 42, 80, 156, 308, 612 (1); Final $x = 612$ (1).	1+1+1
	OR: divisors(104) $= [1,2,4,8,13,26,52,104]$, divisors(65) $= [1,5,13,65]$; Greatest common divisor from the lists $= 13$ (1); Subtraction: $(104,65) \to (65,39) \to (39,26) \to (26,13) \to (13,13) \to (13,0)$, 5 subtractions (1); gcd $= 13$ by both methods (1).	1+1+1
18	Multiples of 3: 11; multiples of 5: 6 (1); Multiples of both (counted twice): 2 (1); Count $= 11+6-2 = 15$ (1).	1+1+1
	OR: Rule: new $x = 2x - 3$ (1); Values: 20, 37, 71, 139 (1); Final $x = 139$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Values of r: 2, 23, 234, 2341 (1); Final $r = 2341$ (2); It reverses the digits (2).	1+2+2
	OR: Sequence: 9, 28, 14, 7, 22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1 (2); Number of steps $= 19$ (3).	2+3
20	Values of r: 7, 71, 712, 7121 (1); Final $r = 7121$ (2); It reverses the digits (2).	1+2+2

	OR: Sequence: 6, 3, 10, 5, 16, 8, 4, 2, 1 (2); Number of steps $= 8$ (3).	2+3

Mistakes to avoid:

- Forgetting to add the carry from the previous column in column addition.
- Counting a carry when the column sum is exactly 9 (a carry needs a sum of 10 or more).
- Leaving out the divisors 1 and the number itself when listing divisors, or listing a divisor twice (as in $36 = 6 \times 6$).
- In Euclid's subtraction method, subtracting the larger number from the smaller or stopping before the two numbers are equal.
- Reading $m \bmod n$ as the quotient of $m \div n$; it is the remainder, and it is always less than n.
- Skipping steps when tracing an algorithm, so that a variable's value is not updated in the table.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 11: **World of Algorithms** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 10	Multiples of 3: 7; of 5: 4; of both: 1.
2	**(b)** 10	The algorithm adds the digits.
3	**(a)** 120	LCM $= \frac{40 \times 15}{5}$.
4	**(d)** 19	$\lfloor \sqrt{389} \rfloor = 19$.
5	**(b)** 7	Pairs: $(572, 312) \to (312, 260) \to (260, 52) \to (208, 52) \to (156, 52) \to (104, 52) \to (52, 52) \to (52, 0)$.
6	**(d)** 9	Pairs: $(306, 243) \to (243, 63) \to (180, 63) \to (117, 63) \to (63, 54) \to (54, 9) \to (45, 9) \to (36, 9) \to (27, 9) \to (18, 9) \to (9, 9) \to (9, 0)$.
7	**(c)** 9	Pairs: $(280, 203) \to (203, 77) \to (126, 77) \to (77, 49) \to (49, 28) \to (28, 21) \to (21, 7) \to (14, 7) \to (7, 7) \to (7, 0)$.
8	**(b)** 7	Pairs: $(304, 176) \to (176, 128) \to (128, 48) \to (80, 48) \to (48, 32) \to (32, 16) \to (16, 16) \to (16, 0)$.
9	**(c)** 21	Multiples of 3: 15; of 5: 9; of both: 3.
10	**(a)** 24	The algorithm adds the digits.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\lfloor \sqrt{133} \rfloor = 11$, so check $2, \ldots, 11$; The smallest divisor found is 7 since $133 = 7 \times 19$ (1); 133 is composite (1).	1+1
12	Replace (m, n) by $(n, m - n)$ re-ordered so the larger is first; $(186, 124) \to (124, 62) \to (62, 62) \to (62, 0)$ (1); 3 subtractions, gcd $= 62$ (1).	1+1
13	$(77, 63) \to (63, 14) \to (49, 14) \to (35, 14) \to (21, 14) \to (14, 7) \to (7, 7) \to (7, 0)$; gcd $= 7$ (1); LCM $= \frac{77 \times 63}{7} = 693$ (1).	1+1
14	$\lfloor \sqrt{193} \rfloor = 13$, so check $2, \ldots, 13$; No number in this range divides it (1); 193 is prime (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	divisors(88) $= [1, 2, 4, 8, 11, 22, 44, 88]$, divisors(40) $= [1, 2, 4, 5, 8, 10, 20, 40]$; Greatest common divisor from the lists $= 8$ (1); Subtraction: $(88, 40) \to (48, 40) \to (40, 8) \to (32, 8) \to (24, 8) \to (16, 8) \to (8, 8) \to (8, 0)$, 7 subtractions (1); gcd $= 8$ by both methods (1).	1+1+1
16	Multiples of 3: 7; multiples of 5: 4 (1); Multiples of both (counted twice): 1 (1); Count $= 7 + 4 - 1 = 10$ (1).	1+1+1
17	Rule: new $x = 2x - 1$ (1); Values: 28, 55, 109, 217, 433, 865 (1); Final $x = 865$ (1).	1+1+1
	OR: divisors(112) $= [1, 2, 4, 7, 8, 14, 16, 28, 56, 112]$, divisors(48) $= [1, 2, 3, 4, 6, 8, 12, 16, 24, 48]$; Greatest common divisor from the lists $= 16$ (1); Subtraction: $(112, 48) \to (64, 48) \to (48, 16) \to (32, 16) \to (16, 16) \to (16, 0)$, 5 subtractions (1); gcd $= 16$ by both methods (1).	1+1+1
18	Multiples of 3: 8; multiples of 5: 5 (1); Multiples of both (counted twice): 1 (1); Count $= 8 + 5 - 1 = 12$ (1).	1+1+1
	OR: Rule: new $x = 2x - 3$ (1); Values: 12, 21, 39, 75 (1); Final $x = 75$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Values of r: 6, 61, 612, 6126 (1); Final $r = 6126$ (2); It reverses the digits (2).	1+2+2
	OR: Sequence: 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1 (2); Number of steps $= 14$ (3).	2+3
20	Values of r: 5, 59, 596, 5964 (1); Final $r = 5964$ (2); It reverses the digits (2).	1+2+2
	OR: Sequence: 12, 6, 3, 10, 5, 16, 8, 4, 2, 1 (2); Number of steps $= 9$ (3).	2+3

Mistakes to avoid:

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- In Euclid's subtraction method, subtracting the larger number from the smaller or stopping before the two numbers are equal.
- Reading $m \bmod n$ as the quotient of $m \div n$; it is the remainder, and it is always less than n.
- Skipping steps when tracing an algorithm, so that a variable's value is not updated in the table.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 11: **World of Algorithms** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 702	LCM $= \frac{26 \times 54}{2}$.
2	**(c)** 8	$\lfloor \sqrt{67} \rfloor = 8$.
3	**(d)** 12	Pairs: $(534, 300) \to (300, 234) \to (234, 66) \to (168, 66) \to (102, 66) \to (66, 36) \to (36, 30) \to (30, 6) \to (24, 6) \to (18, 6) \to (12, 6) \to (6, 6) \to (6, 0)$.
4	**(c)** 229	Pairs: $(458, 229) \to (229, 229) \to (229, 0)$.
5	**(a)** 5	Pairs: $(357, 255) \to (255, 102) \to (153, 102) \to (102, 51) \to (51, 51) \to (51, 0)$.
6	**(b)** 3	Pairs: $(348, 232) \to (232, 116) \to (116, 116) \to (116, 0)$.
7	**(d)** 24	Multiples of 3: 17; of 5: 10; of both: 3.
8	**(d)** 16	The algorithm adds the digits.
9	**(c)** 60	LCM $= \frac{60 \times 15}{15}$.
10	**(b)** 13	$\lfloor \sqrt{193} \rfloor = 13$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Replace (m, n) by $(n, m - n)$ re-ordered so the larger is first; $(228, 152) \to (152, 76) \to (76, 76) \to (76, 0)$ (1); 3 subtractions, gcd $= 76$ (1).	1+1
12	$(111, 87) \to (87, 24) \to (63, 24) \to (39, 24) \to (24, 15) \to (15, 9) \to (9, 6) \to (6, 3) \to (3, 3) \to (3, 0)$; gcd $= 3$ (1); LCM $= \frac{111 \times 87}{3} = 3219$ (1).	1+1
13	$\lfloor \sqrt{110} \rfloor = 10$, so check $2, \ldots, 10$; The smallest divisor found is 2 since $110 = 2 \times 55$ (1); 110 is composite (1).	1+1
14	Replace (m, n) by $(n, m - n)$ re-ordered so the larger is first; $(217, 186) \to (186, 31) \to (155, 31) \to (124, 31) \to (93, 31) \to (62, 31) \to (31, 31) \to (31, 0)$ (1); 7 subtractions, gcd $= 31$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	divisors(84) $= [1, 2, 3, 4, 6, 7, 12, 14, 21, 28, 42, 84]$, divisors(51) $= [1, 3, 17, 51]$; Greatest common divisor from the lists $= 3$ (1); Subtraction: $(84, 51) \to (51, 33) \to (33, 18) \to (18, 15) \to (15, 3) \to (12, 3) \to (9, 3) \to (6, 3) \to (3, 3) \to (3, 0)$, 9 subtractions (1); gcd $= 3$ by both methods (1).	1+1+1
16	Multiples of 3: 6; multiples of 5: 4 (1); Multiples of both (counted twice): 1 (1); Count $= 6 + 4 - 1 = 9$ (1).	1+1+1
17	Rule: new $x = 2x - 3$ (1); Values: 14, 25, 47, 91, 179, 355 (1); Final $x = 355$ (1).	1+1+1
	OR: divisors(72) $= [1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72]$, divisors(45) $= [1, 3, 5, 9, 15, 45]$; Greatest common divisor from the lists $= 9$ (1); Subtraction: $(72, 45) \to (45, 27) \to (27, 18) \to (18, 9) \to (9, 9) \to (9, 0)$, 5 subtractions (1); gcd $= 9$ by both methods (1).	1+1+1
18	Multiples of 3: 13; multiples of 5: 8 (1); Multiples of both (counted twice): 2 (1); Count $= 13 + 8 - 2 = 19$ (1).	1+1+1
	OR: Rule: new $x = 2x - 1$ (1); Values: 13, 25, 49, 97, 193, 385 (1); Final $x = 385$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Values of r: 4, 49, 490, 4905 (1); Final $r = 4905$ (2); It reverses the digits (2).	1+2+2
	OR: Sequence: 7, 22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1 (2); Number of steps $= 16$ (3).	2+3
20	Values of r: 5, 57, 571, 5711 (1); Final $r = 5711$ (2); It reverses the digits (2).	1+2+2
	OR: Sequence: 10, 5, 16, 8, 4, 2, 1 (2); Number of steps $= 6$ (3).	2+3

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